RAG ARCHITECTURE, EXPLAINED

How Sonic Assistant finds an answer

A multi-tenant RAG system where every search is scoped, graph-aware and agentic.

1 **Isolation**
who may read what

2 **GraphRAG retrieval**
vector, keyword and hybrid search plus a knowledge graph

3 **Agentic RAG**
the model decides what to read next

FastAPI · LangGraph · MongoDB Atlas Vector Search · Gemini + Claude

1 · ISOLATION

One scope per request, carried by every read

Sign-in
Clerk identity

↓

Groups map to libraries
a private library each, plus shared ones

↓

`target_scope`
asked for a library you are not in? 403

↓

Vector search
filtered by library

Graph edges
stored per library

Memory
filtered by user

An empty scope reads nothing, never everything

Uploading needs a separate write group

2 · GRAPHRAG RETRIEVAL

Three searches in parallel, then the evidence is checked

Router
a small model decides what the question needs

↓

Chunks
vector, keyword or hybrid

Entity graph
relationships, 2 hops

Session files
priority context

↓

Relevance grader
does this actually answer the question?

Yes
generate, then an evaluator re-checks the answer

No
rewrite the query and retry, 2 passes at most, then an honest fallback

2 · GRAPHRAG RETRIEVAL

Three ways to search, picked by plain rules

No extra model call: the question's wording chooses the strategy.

Vector

USED FOR
everything else

HOW
embed the question, take the nearest chunks by meaning

Lexical

TRIGGERED BY
recent, latest, when, date, current, last

HOW
pull a wide candidate pool, then re-rank by keyword scoring (TF-IDF)

Hybrid

TRIGGERED BY
compare, versus, relationship, how does, why did, or a long question

HOW
run both, then fuse the two rankings (reciprocal rank fusion)

Checked in this order

keyword words first, then comparison words or a long question, otherwise plain vector search

Always inside the scope filter

The strategy changes how chunks are ranked, never which libraries are read.

3 · AGENTIC RAG

When one pass is not enough, the model drives

“Review my last 8 weeks” needs the whole picture, not 3 chunks.

List
what exists in scope

Search
any query, any number of times

Read
whole files, part by part

↓

Decide
answer, or go around again

Example: a 7-week review
list the files, read 5, read the other 2, then answer

- The step budget grows with the size of the library
- The answer names the files it read, and says what it did not
- The same scope rule applies on every tool call

Thresholds measured, writes checked in code